



Nivel 1

Crea una base de datos con MongoDB utilizando como colecciones los archivos adjuntos.

Creo la base de datos “sprint5” usando el botón “+ Create database”.

localhost:27017

>_ Open MongoDB shell

+ Create database

Refresh

Sort by Database Name

View

Luego creo las “collections” usando el botón “+ Create collection”:

localhost:27017 > sprint5

>_ Open MongoDB shell

+ Create collection

Refresh

Sort by Collection Name

View

Create Collection

Collection Name

movies

☐ Time-Series

Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

> Additional preferences (e.g. Custom collation, Clustered collections)

Cancel

Create Collection

Finalmente importo los ficheros .json:

This collection has no data

It only takes a few seconds to import data from a JSON or CSV file.

Import Data

Importo el fichero “movies.json”

My Queries

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

local

sprint5

comments

movies

localhost:27017 > sprint5 > movies

Documents 23.5K

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

EXPORT DATA

UPDATE

DELETE

```

_id: ObjectId('573a1390f29313caabcd4135')
plot: "Three men hammer on an anvil and pass a bottle of beer around."
genres: Array (1)
runtime: 1
cast: Array (2)
num_mflix_comments: 1
title: "Blacksmith Scene"
fullplot: "A stationary camera looks at a large anvil with a blacksmith behind it..."
countries: Array (1)
released: 1893-05-09T00:00:00.000+00:00
directors: Array (1)
rated: "UNRATED"
awards: Object
lastupdated: "2015-08-26 00:03:50.133000000"
year: 1893
imdb: Object
type: "movie"
tomatoes: Object

```

▶

```

_id: ObjectId('573a1390f29313caabcd42e8')
plot: "A group of bandits stage a brazen train hold-up, only to find a determ..."
genres: Array (2)
runtime: 11
poster: "https://m.media-amazon.com/images/M/MV5BMTU3NjE5NzYtYTUyNS00MDVmLWIwYj..."
title: "The Great Train Robbery"
fullplot: "Among the earliest existing films in American cinema - notable as the ..."
languages: Array (1)
released: 1903-12-01T00:00:00.000+00:00
directors: Array (1)
rated: "TV-G"
awards: Object
lastupdated: "2015-08-13 00:27:59.177000000"
year: 1903
imdb: Object
countries: Array (1)
type: "movie"
es: Object

```

Import completed.

23539 documents imported.

Importo el fichero “comments.json”

Compass

My Queries

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- config
- local
- sprint5
 - comments

MongoDB Compass - localhost:27017/sprint5.comments

sprint5

comments

+

localhost:27017 > sprint5 > comments

Documents50.3K

Aggregations

Schema

Indexes1

Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

EXPORT DATA

UPDATE

DELETE

_id: ObjectId('5a9427648b0beebeb69579cc')

name : "Andrea Le"

email : "andrea_le@fakegmail.com"

movie_id : ObjectId('573a1390f29313caabcd418c')

text : "Rem officiis eaque repellendus amet eos doloribus. Porro dolor volupta..."

date : 2012-03-26T23:20:16.000+00:00

>

_id: ObjectId('5a9427648b0beebeb69579cf')

name : "Greg Powell"

email : "greg_powell@fakegmail.com"

movie_id : ObjectId('573a1390f29313caabcd41b1')

text : "Tenetur dorum molestiae ea. Eligendi praesentium unde quod porro. Co..."

date : 1987-02-10T00:29:36.000+00:00

Tenetur dorum molestiae ea. Eligendi
praesentium unde quod porro. Commodi nisi
placeat rerum vero cupiditate neque. Dolorum
nihil vero animi.

_id: ObjectId('5a9427648b0beebeb69579d0')

name : "Talisa Maegyr"

email : "oona_chaplin@gameofthron.es"

movie_id : ObjectId('573a1390f29313caabcd41b1')

text : "Rem itaque ad sit rem voluptatibus. Ad fugiat maxime illum optio iure ..."

date : 1998-08-22T11:45:03.000+00:00

_id: ObjectId('5a9427648b0beebeb69579d3')

name : "Cameron Duran"

email : "cameron_duran@fakegmail.com"

movie_id : ObjectId('573a1390f29313caabcd4217')

text : "Quasi dicta culpa asperiores quaerat perferendis neque. Est animi pari..."

date : 1983-04-27T20:39:15.000+00:00

_id: ObjectId('5a9427648b0beebeb69579d5')

name : "Petry Baelish"

email : "aidan_gillen@gameofthron.es"

movie_id : ObjectId('573a1390f29313caabcd4218')

text : "Quo deserunt ipsam ipsum. Tenetur eos nemo nam sint praesentium minus ..."

date : 2001-07-13T19:25:09.000+00:00

Import completed.
50304 documents imported.

Importo el fichero “sesiones.json”

The screenshot displays the MongoDB Compass application interface. On the left sidebar, under "My Queries", there are icons for zooming out (-), zooming in (+), and a menu (three dots). Below this, the "CONNECTIONS (1)" section shows a search bar labeled "Search connections". A tree view lists the following connections:

- localhost:27017
 - admin
 - config
 - local
 - sprint5
 - comments
 - movies
 - sessions

The "sessions" collection is highlighted. The main panel at the top right shows the breadcrumb path: localhost:27017 > sprint5 > sessions. Below this, tabs for "Documents", "Aggregations", "Schema", "Indexes", and "Validation" are visible, with "Documents" being the active tab. A query input field contains the text: "Type a query: { field: 'value' } or Generate query +". Below the input field are four action buttons: "ADD DATA", "EXPORT DATA", "UPDATE", and "DELETE". The document preview area shows the following JSON data:

```
{
  "_id": ObjectId('5a97f9c91c807bb9c6eb5fb4'),
  "user_id": "t3qulfem@kwiv5.6ur",
  "jwt": "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpYXQiOiE1MTk5MDkzMjEsIm5iZiI6M..."
}
```

A dark green notification banner at the bottom left states: "Import completed. 1 document imported."

Importo el fichero “theaters.json”

My Queries

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

local

sprint5

comments

movies

sessions

theaters

localhost:27017 > sprint5 > theaters

Documents1.6K

Aggregations

Schema

Indexes1

Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

EXPORT DATA

UPDATE

DELETE

_id: ObjectId('59a47286cfa9a3a73e51e72c')

theaterId: 1000

location: Object

_id: ObjectId('59a47286cfa9a3a73e51e72d')

theaterId: 1003

location: Object

_id: ObjectId('59a47286cfa9a3a73e51e72e')

theaterId: 1008

location: Object

_id: ObjectId('59a47286cfa9a3a73e51e72f')

theaterId: 1004

location: Object

_id: ObjectId('59a47286cfa9a3a73e51e730')

theaterId: 1002

location: Object

_id: ObjectId('59a47286cfa9a3a73e51e731')

theaterId: 1010

location: Object

_id: ObjectId('59a47286cfa9a3a73e51e732')

theaterId: 1014

location: Object

_id: ObjectId('59a47286cfa9a3a73e51e733')

theaterId: 1012

location: Object

Import completed.

1564 documents imported.

Importo el fichero “users.json”

My Queries

CONNECTIONS (1)

Search connections

🔍

localhost:27017

admin

config

local

sprint5

comments

movies

sessions

theaters

users

localhost:27017 > sprint5 > users

Document: nikolaj_coster-waldau@gameofthron.es | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

EXPORT DATA

UPDATE

DELETE

```
_id: ObjectId('59b99db4cfa9a34dcd7885b6')
name: "Ned Stark"
email: "sean_bean@gameofthron.es"
password: "$2b$12$UREFwsRUoyF0CrqGNK0Lz00HM/jLhgUCNNIJ9RJAqMUQ74crLj1Vu"
```

```
_id: ObjectId('59b99db4cfa9a34dcd7885b7')
name: "Robert Baratheon"
email: "mark_addy@gameofthron.es"
password: "$2b$12$yGqxLG9LZpXA2xVDhuPnS0Zd.VURVkvz7wg0LY3pn00s7u2S1Z032y"
```

```
_id: ObjectId('59b99db5cfa9a34dcd7885b8')
name: "Jaime Lannister"
email: "nikolaj_coster-waldau@gameofthron.es"
password: "$2b$12$6vz7wiw0.EI5RilvqlzUc./9480gb1uPtXcahDxIadgyC3PS8XCUC"
```

```
_id: ObjectId('59b99db5cfa9a34dcd7885b9')
name: "Catelyn Stark"
email: "michelle_fairley@gameofthron.es"
password: "$2b$12$fiaTH5Sh1zKNFX2i/FTereWGjxoJxvmV7XL.qlfqCr8Cw0xK.mZWS"
```

```
_id: ObjectId('59b99db6cfa9a34dcd7885ba')
name: "Cersei Lannister"
email: "lena_headey@gameofthron.es"
password: "$2b$12$FExjgr7CLhNca.oUsB9seub8mqcHzkJCFZ8heMc8CeIK0zfeTKP8m"
```

```
_id: ObjectId('59b99db6cfa9a34dcd7885bb')
name: "Daenerys Targaryen"
email: "emilia_clarke@gameofthron.es"
password: "$2b$12$NzpbWHdMytemLttFFKduHenr2NZ.rvxIKuYM4AWLTFaUShxbJ.G3q"
```

Import completed.

185 documents imported.

NIVEL I - Ejercicio 1

- Muestra los 2 primeros comentarios que aparecen en la base de datos.

```
> db.comments.find().limit(2)
< {
  _id: ObjectId('5a9427648b0beebeb69579cc'),
  name: 'Andrea Le',
  email: 'andrea_le@fakegmail.com',
  movie_id: ObjectId('573a1390f29313caabcd418c'),
  text: 'Rem officiis eaque repellendus amet eos doloribus. Porro dolor voluptatum voluptates neque culpa molestias. Voluptate unde',
  date: 2012-03-26T23:20:16.000Z
}
{
  _id: ObjectId('5a9427648b0beebeb69579cf'),
  name: 'Greg Powell',
  email: 'greg_powell@fakegmail.com',
  movie_id: ObjectId('573a1390f29313caabcd41b1'),
  text: 'Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Commodi nisi sit placeat rerum vero cupiditate neque.',
  date: 1987-02-10T00:29:36.000Z
}
```

Variación para ver solo los campos de comentarios:

```
> db.comments.find({}, {"text": 1}).limit(2)
< {
  _id: ObjectId('5a9427648b0beebeb69579cc'),
  text: 'Rem officiis eaque repellendus amet eos doloribus. Porro dolor voluptatum voluptates neque culpa molestias. Voluptate unde',
}
{
  _id: ObjectId('5a9427648b0beebeb69579cf'),
  text: 'Tenetur dolorum molestiae ea. Eligendi praesentium unde quod porro. Commodi nisi sit placeat rerum vero cupiditate neque.',
}
```

- ¿Cuántos usuarios tenemos registrados?

`db.users.countDocuments()`

```
> db.users.countDocuments()
< 185
```

- ¿Cuántos cines existen en el estado de California?

`db.theaters.countDocuments({ "location.address.state": "CA" })`

```
> db.theaters.countDocuments({ "location.address.state": "CA" })
< 169
```

- ¿Cuál fue el primer usuario en registrarse?

`db.users.find().limit(1)`

```
> db.users.find().limit(1)
< {
  _id: ObjectId('59b99db4cfa9a34dcd7885b6'),
  name: 'Ned Stark',
  email: 'sean_bean@gameofthron.es',
  password: '$2b$12$UREFwsRUoyF0CRqGnK0Lz00HM/jLhgUCNNIJ9RJAqMUQ74cr1J1Vu'
}
```

- ¿Cuántas películas de comedia existen en nuestra base de datos?

```
db.movies.countDocuments({"genres":{"$in":["Comedy"]}})
```

-- Se usa "\$in" porque el campo es de tipo "Array".

```
> db.movies.countDocuments({"genres":{"$in":["Comedy"]}})
< 7024
```

NIVEL I - Ejercicio 2

Muéstrame todos los documentos de las películas producidas en 1932, pero que el género sea drama o estén en francés.

```
db.movies.find({"year":1932, $or: [{
    "genres": { $in: ["Drama"] },
    "languages": { $in: ["French"]}
}]}
})
```

```
> db.movies.find({"year":1932, $or: [{ "genres":{"$in":["Drama"]}, "languages":{"$in":["French"]} }] })
< {
  _id: ObjectId('573a1392f29313caabcd9a76'),
  plot: 'In this romance a librarian takes a cruise and falls for an unobtainable man, a district attorney married to a crippled woman.',
  genres: [
    'Drama',
    'Romance'
  ],
  runtime: 85,
  rated: 'APPROVED',
  cast: [
    'Barbara Stanwyck',
    'Adolphe Menjou',
    'Ralph Bellamy',
    'Dorothy Peterson'
  ],
  num_mflix_comments: 3,
  poster: 'https://m.media-amazon.com/images/M/MV5BMjEzNTU5NTk2N15BMl5BanBnXkFtZTgwMzA2NTAwMjE@._V1_SY1000_SX677_AL_.jpg',
  title: 'Forbidden',
  fullplot: "On a cruise to Cuba, Lulu Smith falls in love with Bob Grover. Back home, she breaks off the romance when he tells her he is married. Lulu",
  languages: [
    'English',
    'French'
  ],
  released: 1932-01-15T00:00:00.000Z,
  directors: [
    'Frank Capra'
  ],
  writers: [
    'Frank Capra (story)',
    'Jo Swerling (adaptation)'
  ],
  awards: {
    wins: 0,
    nominations: 1,
    text: '1 nomination.'
  }
}
```


NIVEL I - Ejercicio 3

Muéstrame todos los documentos de películas estadounidenses que tengan entre 5 y 9 premios que fueron producidas entre 2012 y 2014.

```
db.movies.aggregate([
  {
    $match: {
      countries: {$in: ["USA"]},
      year: { $gte: 2012, $lte: 2014 },
      "awards.wins": {$gte: 5, $lte: 9}
    }
  }
])
```

```
> db.movies.aggregate([
  {
    $match: {
      countries: {$in: ["USA"]},
      year: { $gte: 2012, $lte: 2014 },
      "awards.wins": {$gte: 5, $lte: 9}
    }
  }
])
< {
  _id: ObjectId('573a13acf29313caabd29366'),
  fullplot: "The manager of the negative assets sector of Life magazine, Walter Mitty, has been working for st
  imdb: {
    rating: 7.4,
    votes: 211230,
    id: 359950
  },
  year: 2013,
  plot: 'When his job along with that of his co-worker are threatened, Walter takes action in the real world e
  genres: [
    'Adventure',
    'Comedy',
    'Drama'
  ],
  rated: 'PG',
  metacritic: 54,
  title: 'The Secret Life of Walter Mitty',
  lastupdated: '2015-08-31 00:10:51.747000000',
  languages: [
    'English',
    'Spanish',
    'Icelandic'
  ],
}
```



Nivel 2

NIVEL II - Ejercicio 1

Cuenta cuántos comentarios escribe un usuario que utiliza "GAMEOFTHRON.ES" como dominio de correo electrónico.

```
db.comments.aggregate([
  {
    $match: {
      email: { $regex: /@gameofthron\.es$/i }           //busca en el dominio
    }
  },
  {
    $group: {
      _id: "$email",                                   // Agrupa por email del comentario
      totalComments: { $sum: 1 }                       // Cuenta cada documento como un comentario
    }
  }
])
```

```
> db.comments.aggregate([
  {
    $match: {
      email: { $regex: /@gameofthron\.es$/i }
    }
  },
  {
    $group: {
      _id: "$email",           // Agrupa por email del comentario
      totalComments: { $sum: 1 } // Cuenta cada documento como un comentario
    }
  }
])
< {
  _id: 'josef_altin@gameofthron.es',
  totalComments: 254
}
{
  _id: 'mark_stanley@gameofthron.es',
  totalComments: 261
}
{
  _id: 'ian_gelder@gameofthron.es',
  totalComments: 267
}
{
  _id: 'charles_dance@gameofthron.es',
  totalComments: 256
}
```

NIVEL II - Ejercicio 2

¿Cuántos cines existen en cada código postal situados dentro del estado Washington DC (DC)?

```
db.theaters.aggregate([
  {
    $match: {
      "location.address.state": "DC"
    }
  },
  {
    $group: {
      _id: "$location.address.zipcode",
      totalTheaters: { $sum: 1 }
    }
  }
])
```

```
> db.theaters.aggregate([
  {
    $match: {
      "location.address.state": "DC"
    }
  },
  {
    $group: {
      _id: "$location.address.zipcode",
      totalTheaters: { $sum: 1 }
    }
  }
])
< {
  _id: '20010',
  totalTheaters: 1
}
{
  _id: '20002',
  totalTheaters: 1
}
{
  _id: '20016',
  totalTheaters: 1
}
```



Nivel 3

NIVEL III - Ejercicio 1

Encuentra todas las películas dirigidas por John Landis con una puntuación IMDb (Internet Movie Database) de entre 7,5 y 8.

```
db.movies.aggregate([
  {
    $match: { directors: "John Landis",
              "imdb.rating": {$gte:7.5, $lte:8}
            }
  }
])
```

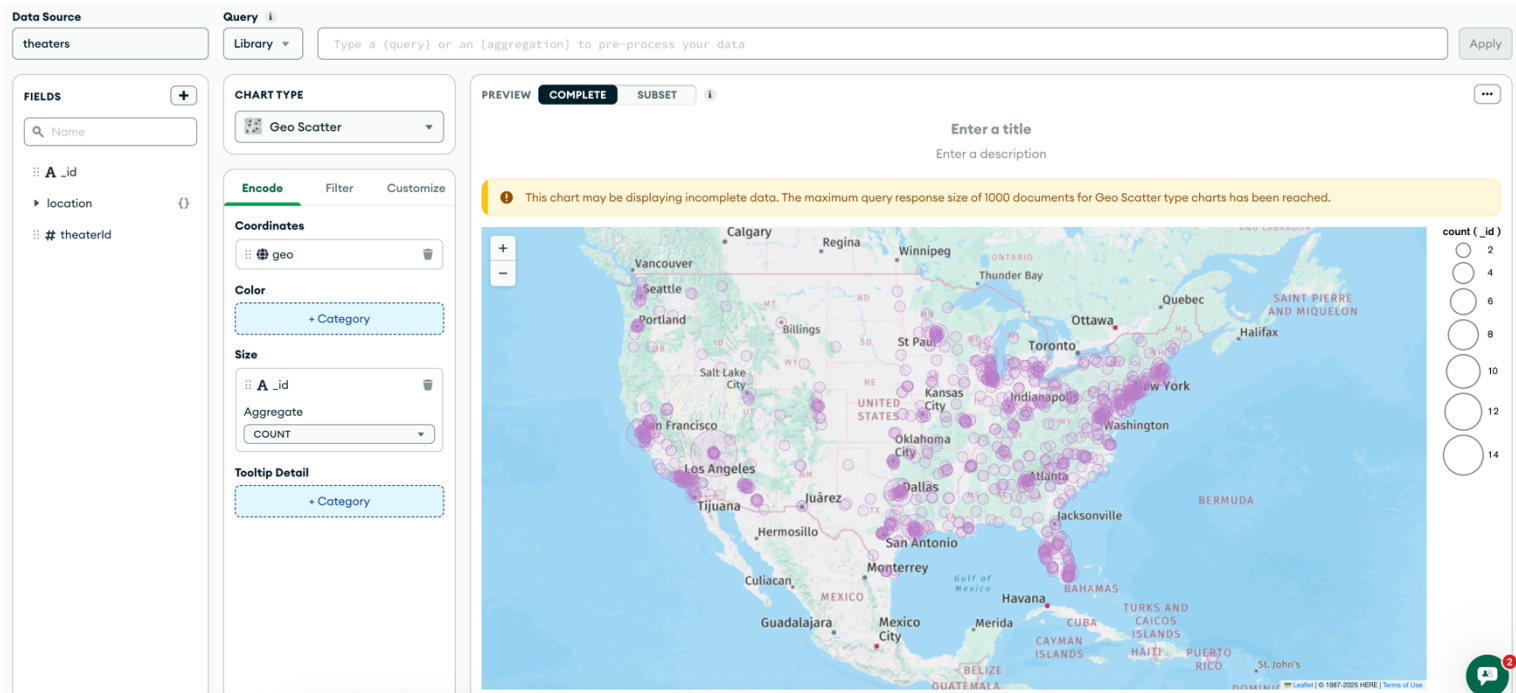
```
> db.movies.aggregate([
  {
    $match: { directors: "John Landis",
              "imdb.rating": {$gte:7.5, $lte:8}
            }
  }
])
< {
  _id: ObjectId('573a1397f29313caabce6d94'),
  fullplot: "Faber College has one frat house so disreputable it will take anyone. It has a second one full of white, and",
  imdb: {
    rating: 7.6,
    votes: 84834,
    id: 77975
  },
  year: 1978,
  plot: 'At a 1962 college, Dean Vernon Wormer is determined to expel the entire Delta Tau Chi Fraternity, but those troo',
  genres: [
    'Comedy'
  ],
  rated: 'R',
  metacritic: 82,
  title: 'Animal House',
  lastupdated: '2015-09-13 00:02:47.803000000',
  languages: [
    'English',
    'Italian'
  ],
  writers: [
    'Harold Ramis',
    'Douglas Kenney',
    'Chris Miller'
  ],
  type: 'movie',
```

NIVEL III - Ejercicio 2

Muestra en un mapa la ubicación de todos los teatros de la base de datos.

Primero me registre en MongoDB, crear un cluster de datos en algún servidor de bases de datos (AWS, Google, etc). Luego vinculo mi base de datos de MongoDB Compass a esta instancia web.

Finalmente con la herramienta “Chart” genero el mapa solicitado:



Hay un warning que limita los datos visualizados a 1000, debido a que la collection “theaters” tiene 1564 documentos.

```
> db.theaters.countDocuments()  
< 1564
```